AI PRODUCT CASE STUDY · B2C / QUICK COMMERCE

The Usual, Done Right

How I would build an AI shopping and reorder assistant for 10-minute delivery in India and grocery delivery in the US

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The short version

- **The bet:** the winning grocery assistant is not the chattiest one. It is the one that gets your usual basket right in one tap, understands "2 packet milk aur aadha kilo tamatar" as well as "high-protein dinners for two", and never sneaks anything into the cart.
- **Two markets, two products.** India: ₹691 baskets[2], 10-minute delivery, code-mixed voice, razor-thin margins. US: $115 baskets[5], weekly planning, substitutions decided mid-pick.
- **North star:** kept assisted baskets per week, not basket size. Pushing AOV with AI is how you end up on the wrong side of India's ban on basket sneaking[12].
- **Hardest constraint:** cost. Blinkit earned 0.6% adjusted EBITDA on order value in Q1 FY27[3]. On a ₹691 basket that is roughly ₹4, so my AI budget is ₹1 per assisted order.

1. Why this problem

I picked grocery because it is the most repeated purchase most of us make, and the experience still asks people to rebuild the same list from scratch. Search, scroll, add, repeat, every few days.

Two things changed over the last year. Assistants started showing real commerce results: Amazon says Rufus drove about $12 billion in incremental annualized sales in 2025[7], and Instacart says orders placed with its assistant are larger than its $115 average basket[5]. And in India, the platforms started building for how people actually talk: Swiggy partnered with Sarvam for voice ordering in 11 Indian languages[9], and Zepto rebuilt search with custom language models so "kapde rakhne ki bag" finds a laundry basket[10].

So the question is no longer "should grocery have an AI assistant". It is: what should it do first, how do you know it is working, and how do you stop it from becoming a smarter upsell machine.

2. Same job, two very different markets

The job to be done is identical everywhere: keep my kitchen stocked and help me decide what to cook, with as little effort as possible. Almost everything around that job is different.

In the table below, the row I keep coming back to is substitution. In India the platform owns the inventory and knows what is out of stock before you pay, so the assistant can fix the basket up front. In the US the substitution moment is in the aisle, mid-order, with a human shopper. Same feature name, completely different product.

	India (Blinkit, Zepto, Instamart)	US (Instacart, DoorDash, Amazon Fresh)
Basket	Small and frequent. Instamart AOV ₹691 in Q1 FY27[2]	Large and weekly. Instacart AOV about $115[5]
Speed	10 to 15 minutes from a dark store	Same day, often within an hour
Assortment	One dark store, a few thousand SKUs, stock known before checkout	~100,000 stores, 2B+ catalog items[6], stock uncertain until a shopper picks
When substitution happens	Before payment, on screen	During picking, shopper and customer chat
Language	Code-mixed: Hinglish, Tenglish. 57% of urban users prefer Indic-language content[11]	Mostly English, some Spanish
Margin room for AI	Very thin. Blinkit adj. EBITDA 0.6% of NOV[3]	More room per order; latency matters more than cost
Regulation that bites	CCPA dark-pattern rules[12], DPDP Act and Rules[14]	FTC Section 5 and dark-pattern scrutiny[15], state privacy laws, FDA allergen labelling[16]

3. Who I am building for

Lakshmi, 58, Kukatpally (Hyderabad)

Runs a joint-family kitchen. Comfortable speaking Telugu, slower typing in English. Today her son places orders for her. Wants: say what she needs, see it, done. Pain: search does not understand "kandi pappu" or "chinna packet".

Arjun and Meera, 29 and 31, Gachibowli

Working couple, orders three or four times a week, mostly top-ups. Wants: one tap for the usual morning basket. Pain: rebuilding the same 12-item cart, and finding out at checkout that the paneer is out of stock.

Jess, 38, Columbus, Ohio

Two kids, one weekly order around $150. Plans meals on Sunday night. Wants: "five easy dinners, one vegetarian, under $120" turned into a cart. Pain: substitutions that ignore her son's nut allergy.

Dev, 26, San Francisco

Single, orders snacks and essentials on DoorDash between big shops. Wants: fast reorder and smart restock reminders. Pain: notifications that feel like ads.

I would start with Arjun and Meera in India and Jess in the US. They have the highest order frequency and the clearest repeat patterns, which is exactly where a reorder model earns trust fastest. Lakshmi is the bigger long-term unlock in India, but she needs voice to be very good before she should see it.

4. Market and evidence

India. Redseer puts quick commerce at roughly $13 to 14 billion in FY26, about 17% of India's ~$80 billion online retail and growing around 120% a year; it is now about 70% of online grocery[1]. Blinkit alone did ₹17,132 crore of net order value in Q1 FY27, up 86% year on year, across 2,443 stores[3]. Instamart averaged 13.5 million monthly transacting users and delivered 114.5 million orders in the same quarter[2].

US. Online grocery hit a record $12.7 billion in December 2025, 19% of all grocery spend[4]. Instacart alone processed $10.35 billion of GTV and 90.3 million orders in Q2 2026[5].

What assistants have shown so far. Walmart says customers who use Sparky have an average order value about 35% higher than those who do not[8]. Instacart reports assistant orders carry more items than its typical basket, and that customers start with restocking before moving to recipes[6].

> My read on that evidence: "users of the assistant spend more" is a selection effect until proven otherwise. Heavy shoppers try new features first. Before I claim any lift, I want a randomized holdout. It is the first thing I would set up, before the first model.

Sizing the opportunity (illustrative, my assumptions). Take an Instamart-scale platform at 114.5 million orders a quarter[2]. If 10% of orders become assistant-built in year one and each carries one extra kept item worth about ₹60, that is around ₹69 crore of extra GOV per quarter. At Instacart scale, 10% of 90.3 million orders[5] at +$8 each is roughly $72 million of GTV a quarter. Both are estimates. The bigger prize in India, in my view, is frequency and retention, not basket size, and that is how I would frame the business case.

5. Why AI here, and where not

Not every part of this needs a large language model. Putting one everywhere would blow the cost budget and add failure modes. Here is my split.

Job	Approach	Why
Predict what you need this week	Classic ML	Purchase history plus time since last buy. Cheap, fast, explainable, and it is our data moat.
Understand free-form voice or text	LLM + speech-to-text	Code-mixed language, units like "aadha kilo", vague asks like "breakfast stuff". Rules cannot keep up.
Turn a recipe or occasion into a basket	LLM, grounded in catalog	Needs reasoning about quantities and dietary constraints.
Match words to real products	Retrieval + rules	Embedding search with a vernacular synonym layer, then a hard check that the SKU is in stock.
Pick substitutes	ML ranking + hard rules	Learned preferences, but allergens, veg/non-veg and EBT eligibility are never left to a model.
Price, offers, final total	Deterministic	Money is never generated text.

6. The product: five core flows

1. The usual, in one tap. A home-screen card: "Your Tuesday basket, 12 items, ₹840." Built from the reorder model, already checked against the store's live stock. Out-of-stock items show a suggested swap that the user must accept. This is the MVP and the trust builder.

2. Say it. Tap the mic and speak in Telugu, Hindi, English or any mix. "Doodh do packet, bread, aur Sunday ke liye chicken." The assistant shows the basket on screen instead of reading it back. Showing is faster, cheaper and easier to correct than speaking.

3. Cook something. "Paneer butter masala for four" or "five easy dinners under $120." The assistant builds the ingredient list, skips what you bought recently, and marks every item it added so you can remove it.

4. Fix it before it breaks. India: stock-outs resolved before payment with a ranked swap. US: pre-approved substitution rules ("same brand any size", "never substitute") and a smarter suggestion when the shopper is in the aisle. About 90% of online grocery shoppers say there is at least one attribute they will not compromise on in a substitute[18], so these rules are set per item, not per account.

5. Snap a list. Photograph a handwritten list. Instacart already offers this in the US[6]. In India I would test it with households where one person writes and another orders.

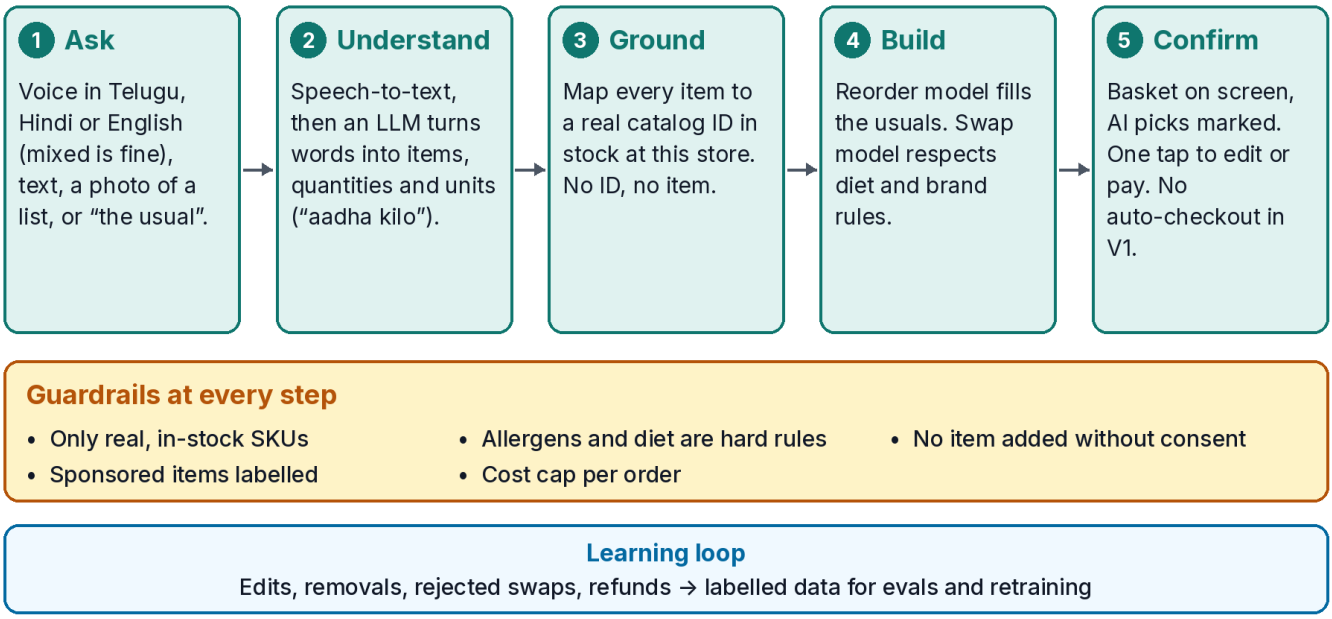


Figure 1. The assistant's core loop and the guardrails around it.

7. The AI approach

Build vs buy

My rule: buy the commodity, build the moat. Speech-to-text and the base language model are commodities that improve every quarter. The purchase graph, the store's live inventory and the substitution feedback are ours, and nobody can buy them.

- **Buy:** speech-to-text tuned for Indian languages (I would bake off an India-first provider such as Sarvam against global options on our own audio), and a hosted LLM for parsing, with structured JSON output constrained to catalog IDs.
- **Build:** the reorder model, the substitution ranker, the catalog retrieval layer with vernacular synonyms, and the guardrail service.
- **Later:** distill a small in-house model once we have a few hundred thousand labelled parses. That is the main lever for cost in India.

Data I need

- Order history at item level, with timestamps, to model replenishment cycles.

- Search logs, especially null and low-result queries. Those are the vocabulary the assistant has to learn.
- Substitution outcomes: accepted, rejected, refunded. The single best training signal we have.
- A labelled voice set per language: real code-mixed audio, kitchen noise, older speakers. Collected with explicit consent under DPDP[14].

From my GenAI work on voice analytics for customer and surveyor calls at ICICI Lombard, I know how messy real Indian audio is. A model that scores well on clean benchmark audio can still fall apart on a phone call from a busy street. I would never trust a vendor's word error rate; I would trust ours.

Human in the loop

In V1 the customer is the human in the loop. The assistant proposes, the customer confirms. There is no auto-checkout. Behind the scenes, an annotation team reviews low-confidence parses every week in each launch language, and in the US the shopper stays in the loop for substitutions. Instacart runs a network of about 600,000 shoppers[5], and I would treat them as a feedback source, not just labour.

Evals and the quality bar

These are my launch bars. They are targets I would set and defend, not industry figures.

What I test	How	Launch bar (my target)
Voice and text parsing	Golden set of 2,000 real utterances per language, incl. code-mixed and noisy audio	Item-level F1 ≥ 90% on top 500 SKUs; quantity and unit accuracy ≥ 95%
Reorder prediction	Backtest against each household's actual next basket	Beats the "repeat last order" baseline on precision@10
Substitutions	1,000 pairs rated by trained raters	≥ 85% rated acceptable; hard-rule violations = 0
Grounding	Automated check on every response	0 items without a valid, in-stock catalog ID
Online	A/B test with a long-running 5% holdout	No guardrail metric worse than control

Running six ML fraud models in production at ICICI Lombard taught me that the eval set is a product, not a one-time task. It needs an owner, a refresh cadence and a drift alarm, or it slowly stops describing reality.

Latency and cost

Latency: p95 under 3 seconds from text to a ready basket, under 5 seconds for voice. For a 10-minute delivery promise, a 15-second assistant feels broken.

Cost in India: Blinkit's 0.6% margin[3] applied to Instamart's ₹691 basket[2] leaves roughly ₹4 of profit per order. My budget for AI is ₹1 per assisted order. Published speech-to-text pricing from one Indian provider is ₹0.41 to 0.49 per minute[17], so a 30-second voice request costs about ₹0.25 in speech alone. LLM parsing on a small model should add well under ₹0.50 (my estimate, to be confirmed in a bake-off). That only works if the one-tap reorder path uses no LLM at all, voice replies are shown rather than spoken, and repeated parses are cached.

Cost in the US: a $115 basket gives more room. I would spend it on a richer model for meal plans, capped at about $0.05 per assisted order.

Failure modes and guardrails

What can go wrong	What I do about it
Mishears "dahi" as "daal", or "1 kg" as "1 packet"	Show, do not say: the basket is on screen with quantities editable. Low-confidence items are flagged.
Suggests a product the store does not have	Output is constrained to in-stock catalog IDs. No ID, no item.
Swaps in something with an allergen	Allergens, veg/non-veg and EBT eligibility are hard filters, outside the model. Sesame is the ninth major US allergen[16]; the list lives in code, not a prompt.
Quietly inflates the basket	Every AI-added item is marked and needs a tap. No pre-ticked add-ons. This is also the law in India[12].
Ads leak into "helpful" suggestions	Sponsored items labelled, capped, and ranked separately from organic suggestions.
Ordering on someone's behalf goes wrong	Household profiles with per-member diet rules, plus an order summary sent to the account owner.

Responsible AI and regulation

India. The CCPA's 2023 dark-pattern guidelines ban 13 patterns, including basket sneaking, false urgency and drip pricing[12]. Basket sneaking is defined as adding items at checkout without the user's consent so the total goes up. An assistant that "helpfully" adds things is one bad design review away from that. In 2025 the CCPA asked platforms to self-audit, and Zepto, Swiggy and Blinkit all filed declarations[13]. I would bake the rules into the PRD: consent per added item, "only 2 left" shown only if the inventory says so, and an audit log of every assistant action. On data, the DPDP Rules were notified on 14 November 2025 with an 18-month phased compliance window[14]. Voice recordings are personal data, so I would collect them with a separate, plain-language consent and keep them only as long as the eval set needs them.

US. The FTC Act bars unfair or deceptive practices, and the FTC laid out its view of dark patterns in a 2022 staff report[15]. The same consent-per-item design covers it. State privacy laws give users rights over their personal data, voice included, so delete and opt-out have to work from day one.

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8. How I would measure success

I chose "kept assisted baskets per week" as the north star because it rewards the assistant only when it saves effort and the customer agrees with what it did. Basket size is an outcome I watch, not the target. If I optimise an AI for AOV, I have built a very polite upsell engine, and in India a legally risky one.

Definition: an order where the assistant built at least half the items, and none of the assistant-chosen items were removed after delivery, refunded or complained about.

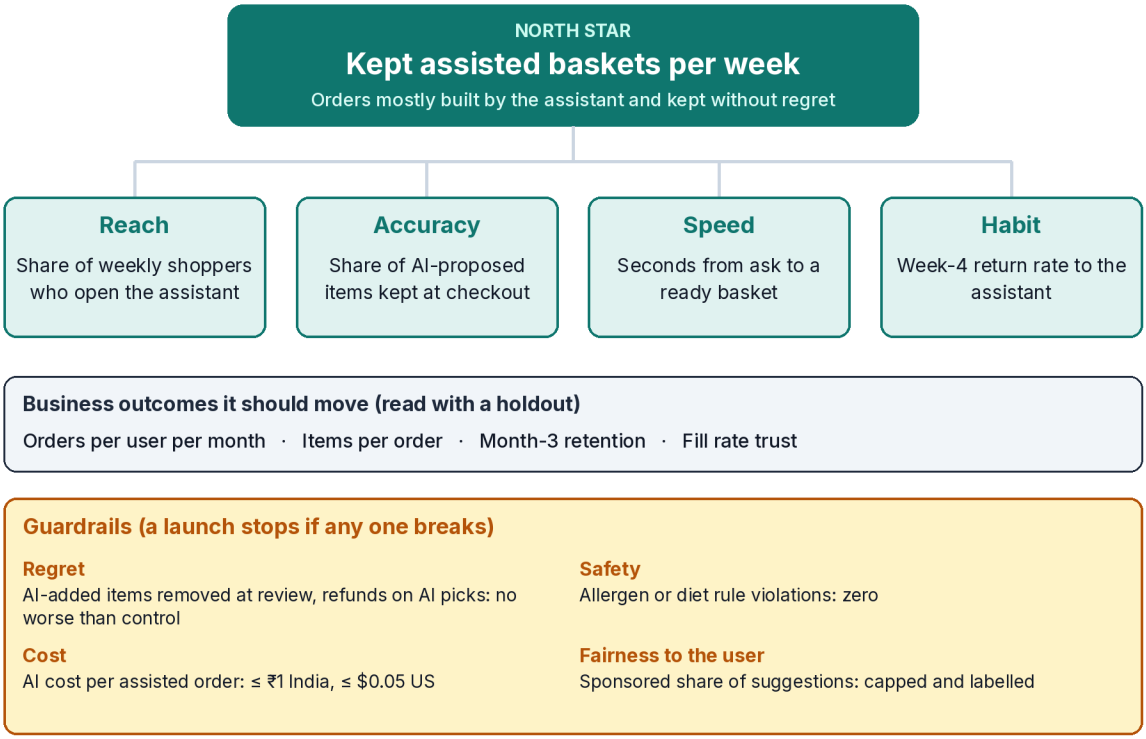


Figure 2. Metrics tree: north star, input metrics, outcomes and guardrails.

9. MVP and roadmap

Phase	Scope	What it proves	Exit gate
MVP (0 to 3 months)	"The usual" card and pre-checkout swaps. Text only. Hyderabad (India), one metro (US).	People trust a predicted basket	Item keep rate ≥ 70%; no guardrail breach
Phase 2 (3 to 6 months)	Voice in Telugu, Hindi and English, recipe-to-basket, US meal plans	AI understands real requests	Parsing bars met; cost ≤ ₹1
Phase 3 (6 to 12 months)	Opt-in auto-replenish with price cap and edit window, household profiles, more cities and languages	Habit, at scale	Holdout shows lift in frequency or retention

What I am deliberately not building first: a full chat interface. People start with restocking[6], and a chat box makes the user do the explaining. A pre-built basket does the work for them.

10. Go-to-market

India. Launch inside the app, not as a separate product. The "usual" card sits on the home screen for households with four or more orders a month. Voice goes into the search bar as a mic, starting in Hyderabad, where Telugu, Hindi and English mixing is the norm. Opt-in WhatsApp restock nudges come later, only for users who ask for them. The pitch to users is time, not tech: "Your usual, ready in one tap."

US. Lead with Sunday planning. "Plan my week" on the home screen on weekends, connected to loyalty pricing and EBT-eligible filters. For retailers, the same engine can ship as a white-label assistant, the route Instacart is taking with Cart Assistant[6].

Where I would position against competitors: Swiggy has moved first on Indian-language voice with Sarvam[9], and Instacart has a strong head start in the US[6]. I would not try to out-chat them. I would compete on reorder accuracy and on never getting a substitution wrong, because those are what bring people back.

11. Risks and how I would de-risk them

Risk	Severity	How I de-risk it
Voice accuracy too low for older or non-English speakers	High	Golden sets per language before launch; text-first MVP; voice released language by language only when it clears the bar.
AI cost eats a 0.6% margin	High	Classic ML for reorder; LLM only for free-form input; distill a small model in Phase 2; cost per order on the daily dashboard.
Regulator or press calls it basket sneaking	High	Consent per AI-added item, audit log, legal review in the PRD, guardrail metric on removals.
One bad substitute breaks trust	Medium	Per-item rules, hard allergen filters, instant credit when a swap is rejected at the door.
Lift is not real, only selection	Medium	5% holdout from day one; decisions made on holdout deltas only.

From my work scaling DIY vehicle inspection at ICICI Lombard, where adoption grew 30%, my lesson was that self-serve AI wins on trust, not cleverness. The same applies here: every guardrail above is also a growth feature.

12. What I would measure in the first 90 days

- **Days 1 to 30, learn:** what share of each household's basket is a repeat (I would measure this, not assume it), the top null-search queries by language, and today's substitution rejection rate. Build the golden sets and the holdout.
- **Days 31 to 60, prove trust:** ship "the usual" to 1% of eligible users in Hyderabad and one US metro. Track item keep rate, time to checkout and the removal guardrail, every day.
- **Days 61 to 90, decide:** ramp to 10% only if keep rate is at least 70% and no guardrail is worse than control. Read week-4 return rate and orders per user against the holdout. Close with a one-page scale, fix or stop memo, against thresholds agreed with engineering and legal before day 1.

> If I had to pick one number to watch on day 90: the share of AI-proposed items customers keep. If people keep what the assistant picks, everything else follows. If they do not, no amount of conversation will save it.

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Notes. Estimates and my own targets are labelled in the text: the opportunity sizing in section 4, the LLM cost per request, the ₹4 profit-per-order proxy (one company's margin applied to another's basket), and every launch bar and cost cap. Personas are illustrative composites, not real customers. Figures for other companies are as those companies or the cited outlets reported them. This is a self-initiated case study, not a product I shipped; the only employer work referenced is my own ICICI Lombard experience, named as such.